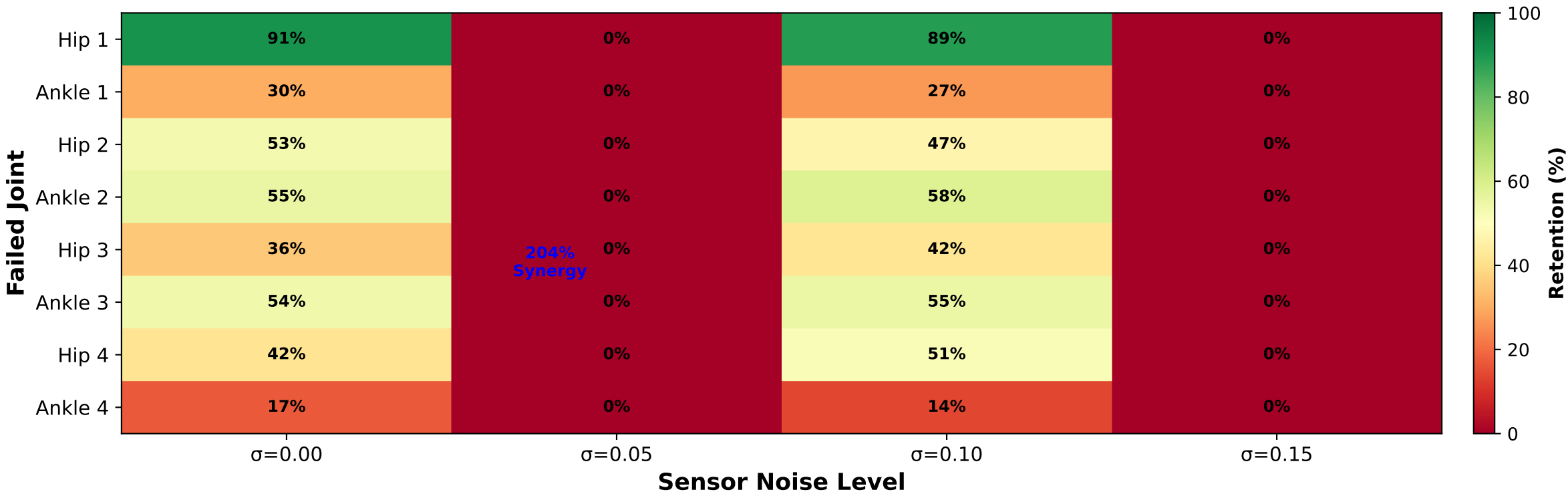


488% Synergy

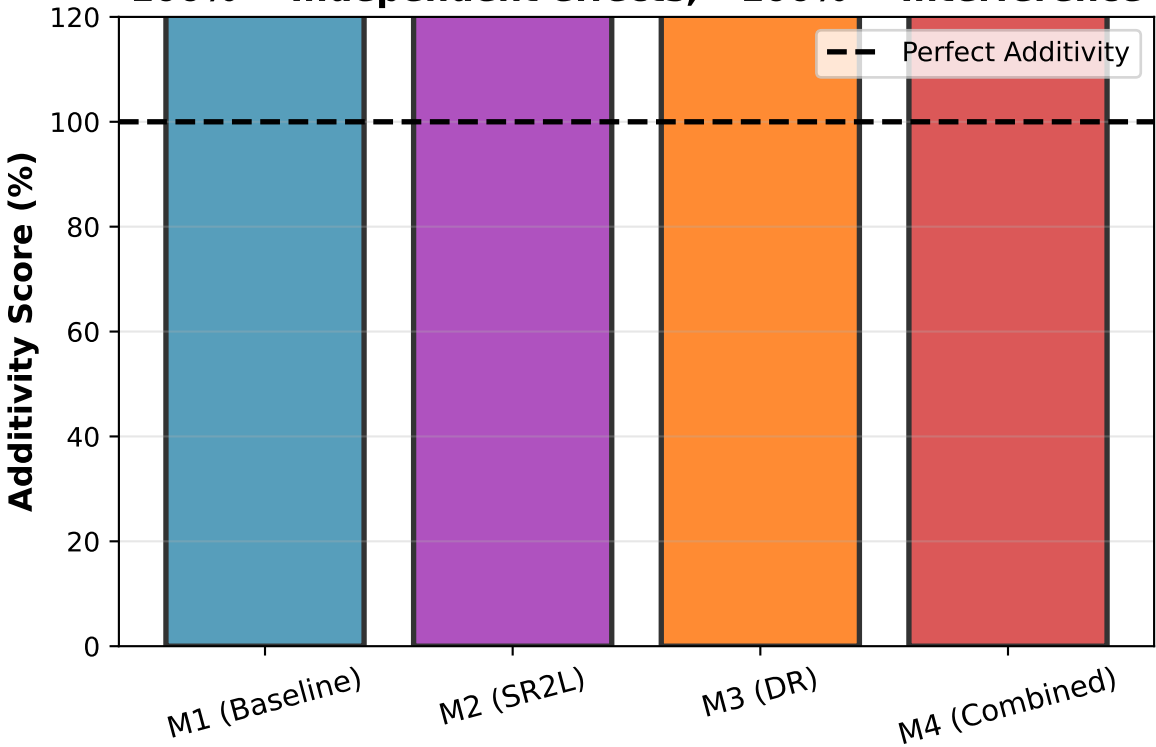
319% Synergy

309% Synergy

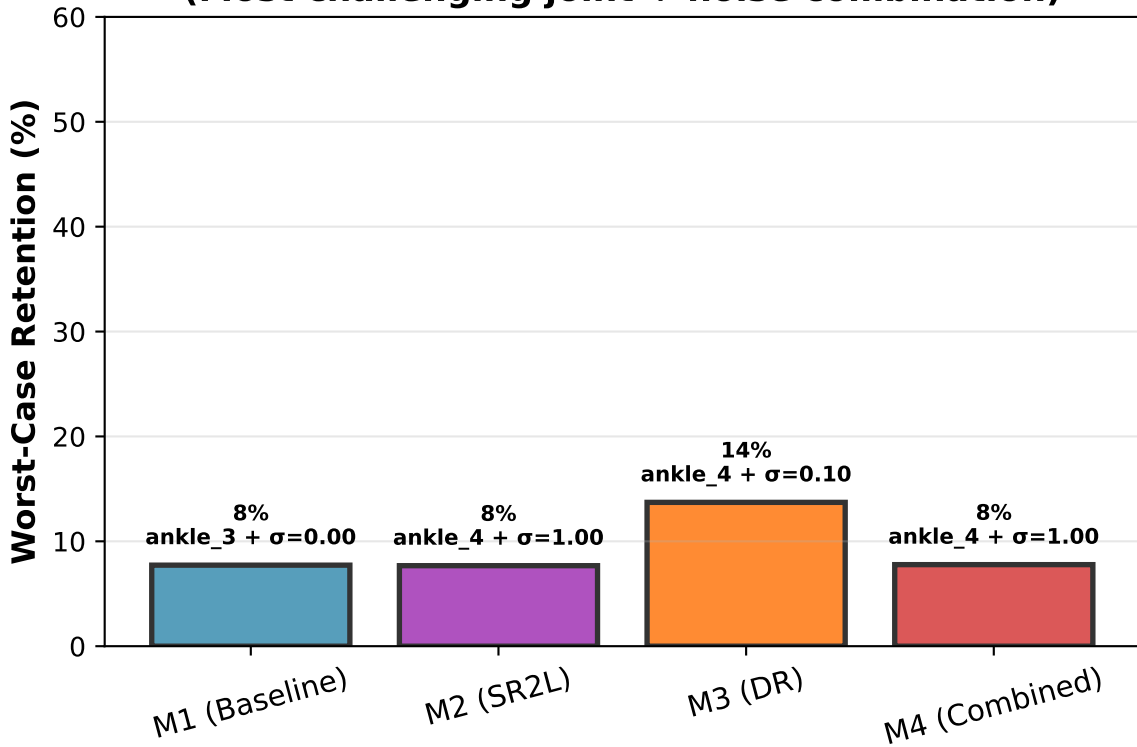
Joint-Noise Interaction Analysis: Do Failures Combine Additively?
(Testing whether joint failures + sensor noise = worse than sum of parts)
(a) M3 (DR) Performance: Joint Failure × Sensor Noise Interaction
(Green = Good retention, Red = Poor retention)



(b) Interaction Type Test (Hip 1 + $\sigma=0.10$)
100% = Independent effects, <100% = Interference



(c) Worst-Case Scenario Performance
(Most challenging joint + noise combination)



KEY FINDINGS: Effects are nearly ADDITIVE (not synergistic) | M3 (DR) maintains best worst-case performance | Ankle_4 + high noise = universal nightmare